

Lab 7 - Rotating Penny

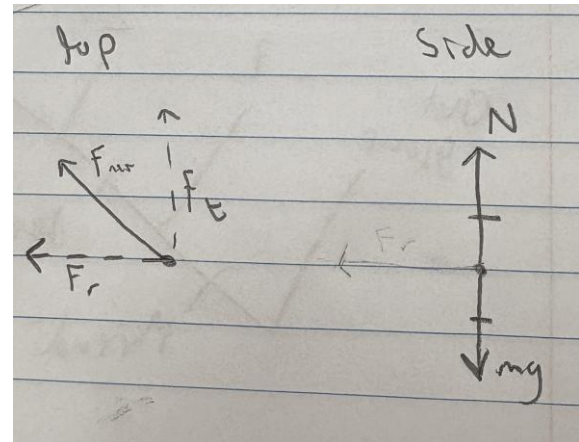
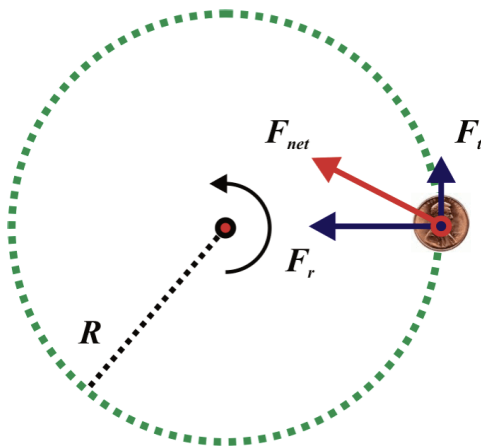
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Learning Outcomes

- To recognize that the relationships between angular position θ , angular velocity ω and angular acceleration α are analogous to their linear counterparts position x , velocity v and acceleration a .
- To apply Newton's laws to the forces acting on a penny on a surface that is accelerating in non-uniform circular motion (the normal force, gravity force and friction force).
- To identify the components of friction keeping a penny moving in non-uniform circular motion. (the radial component changes the direction of the penny; the tangential component the speed).
- To determine the magnitude of the total acceleration of an object in non-uniform circular motion and identify the condition for sliding for the penny.

Pre-lab (background)

Radial forces will cause an object to move in a circle, while **tangential** forces will cause that object to speed up or slow down along its circular path. Since centripetal means “center seeking,” a centripetal force is also a radial force. These words and subscripts (F_c , F_r) can be used interchangeably.



top-view FBD

side-view FBD

Consider a penny is at rest relative to a point on the rotating surface on which it lies. For the penny to remain at rest on the surface, a net force must act on the penny to cause it to move in circular motion with the surface. The **net force is the static frictional force** between the penny and the surface of the turntable. The net frictional force has 2 orthogonal components: radial (F_r or F_c) and tangential (F_t).

1. Insert a **top-view Free Body Diagram** for the penny next to the figure. Represent the net force of static friction as its two orthogonal components.
2. Insert a **side-view** FBD for the penny next to the top-view FBD and identify the balanced forces.
3. Write Newton's second law as one vector equation:

$$F = m(a_c + a_t)$$

4. Write Newton's second law as 3 equations in 3 orthogonal directions and remember the equations for centripetal acceleration $a_c = v^2/r$ and tangential acceleration $a_t = \alpha r$

In r (radial horizontal):

$$F_r = ma_c$$

$$F_r = m \frac{v^2}{r}$$

In t (tangential horizontal):

$$F_t = ma_t$$

$$F_t = m\alpha r$$

In y (vertical):

$$F_y = ma_y = 0$$

$$N - mg = 0$$

$$N = mg$$

5. Write the equation for the magnitude of the total acceleration $\mathbf{a}_{\text{total}}$ vector of the penny in non-uniform motion in terms of its components $\mathbf{a}_c$ and $\mathbf{a}_t$:

$$a_{\text{total}} = \sqrt{a_c^2 + a_t^2}$$

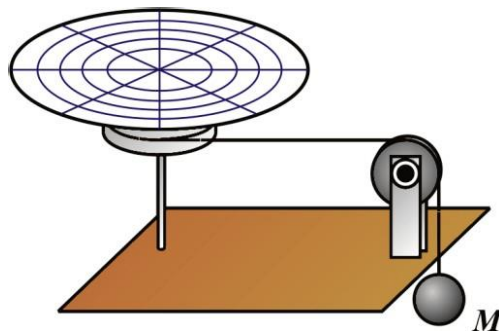
6. Write the equation for the magnitude of the maximum static friction $\mathbf{F}_{s\text{max}}$ in terms of the normal force F_n

$$F_{s,\text{max}} = \mu_s N$$

$$F_{s,\text{max}} = \mu_s mg$$

Materials

- Laptop with Capstone
- Rotary motion sensor
- Turntable
- Pulley
- String
- Penny
- Calipers
- Ruler
- Base and rod
- 10-gram mass hanger



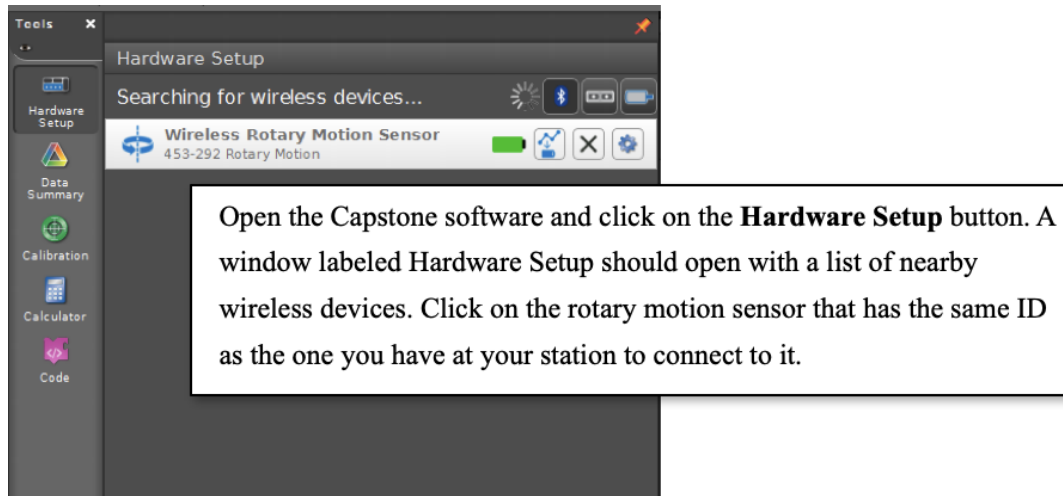
Set up (check off what is complete)

- Mount the rotary motion sensor on a support rod such that the 3-step pulley is on top and able to rotate freely.
- Mount the clamp on super pulley to the narrow side of the rotary motion sensor, as shown in the picture on the right.
- Cut a piece of string to about a length of 1.5 m. Tie one end of the string to the large groove of the rotary sensor's 3-step pulley. This can be done by threading the string through one of the small holes in the pulley and tying a knot.

- ~~Drape the thread over the super pulley and tie its other end to the mass hanger. Adjust the angle of the super pulley such that the thread is in the middle of the super pulley's groove and tangent to the step pulley.~~
- ~~Attach the turntable platform to the top of the 3-step pulley, as shown in the picture on the right.~~

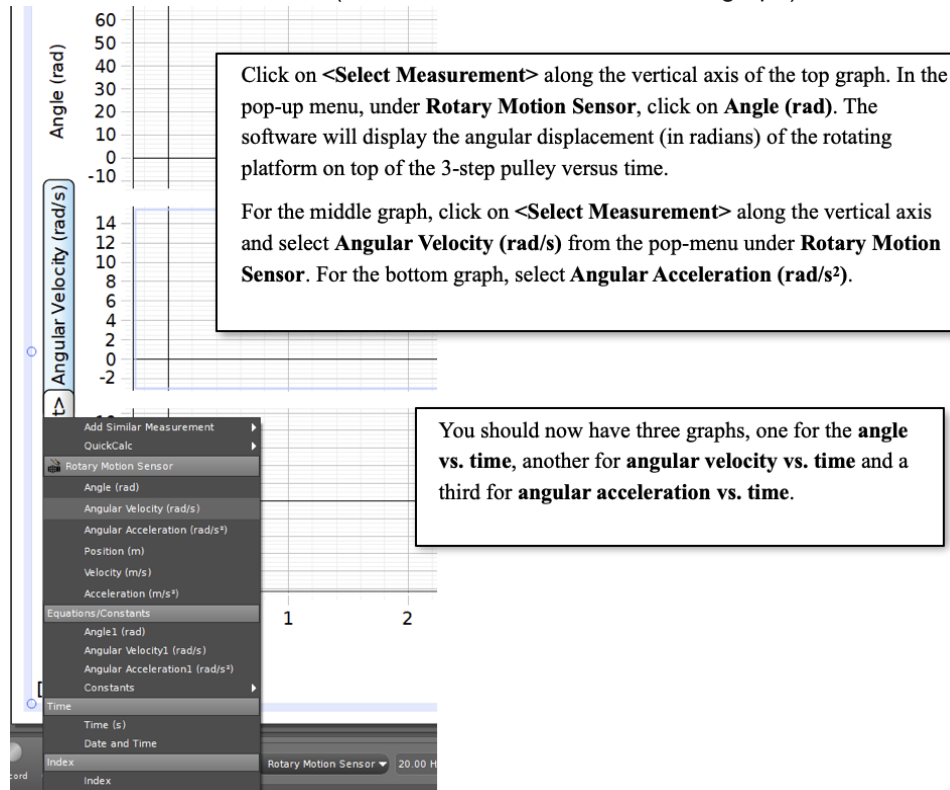
Activity 1: rotational kinematics

The purpose of this first activity is to develop your understanding of the relationships between angular position θ , angular velocity ω and angular acceleration α by analyzing the data from three graphs of rotational motion of a turntable starting from rest and accelerating with constant angular acceleration. And to recognize they are analogous to their linear counterparts position x , velocity v and acceleration a .



1. Click on the **Hardware Setup** button again to close the Hardware Setup window.
2. Double-click on the **Graph** icon to open a new graph.

- Click on the icon with two axes and a green gear twice to add two additional graphs that each **share the same time axis** (located in the toolbar above the graph).



4. Attach the hanging mass ($M=10$ grams) to the end of the string, as shown in the picture on the previous page, and wind the string around the 3-step pulley. Hold the turntable still.

5. Press the **Record** button and allow the mass to fall by releasing the turntable. Press **Stop** before the mass hits the floor. Repeat until you have 3 recognizable graphs.

6. **Compare** the three graphs and explain how they are related in terms of shape and slope:

The θ vs. time trendline shape is... Slowly increasing in a exponential shape positive

The ω vs. time trendline shape is... A constant linear shape that is positive

The α vs. time trendline shape is... Our acceleration is constant linear flat line.

The θ vs. time graph and the ω vs. time graph are related to each other by... Them both increasing a similar times, and θ vs. time graph is the slope of the graph while the ω vs. time graph is the area under the graph

The ω vs. time graph and the α vs. time graph are related to each other by...

The ω vs. time graph shows the velocity of the experiment while the α vs. time graph represents the rate of change of the ω vs. time graph.

7. How do these observations remind you of **x vs. time**, **v vs. time** and **a vs. time graphs** in chapter 2?

X vs time graphs in chapter 2 were similar to these graphs as they tended to be linear and positive, similar to the angle and angular velocity graphs.

A vs. Time graphs from chapter two were similar to the angular acceleration as they tended to be flat, meaning a constant acceleration.

8. Use the graphs to measure the angular acceleration α . Describe the method you used. Could you use a second method to check your result? Explain.

We used the capstone to find the average angular acceleration from the three trials at 0.098 rad/s^2 .

Activity 2: the coefficient of static friction

In this activity, you will use the results of a penny slipping off of the turntable at a particular radius to determine the coefficient of static friction between the surface of the turntable and a penny.

1. Carefully rewind the string around the base of the turntable and place the string over the pulley with the hanging mass attached. The hanging mass must be great enough to cause the penny to slip at several radii (5-10 cm), but small enough to keep it from slipping too quickly. (Start with $M = 10 \text{ g}$).
2. **Use calipers** to place the penny at a radial distance $r_1 = 5 \text{ cm}$ from the center of the turntable.
3. Press **Record** and release the turntable from rest. **Use your hand to stop the turntable at the moment the penny begins to slide from its initial position.** Press **Stop** to end data collection. Repeat this step if there are any artifacts (spikes) in the graphs.
4. Use the [googlesheet template](#) to record the following measurements at the moment just prior to when the penny slid from $r = 5 \text{ cm}$:
 - a. magnitude of ω , 8.884 rad/s^2
 - b. magnitude of α , 1.327 rad/s^2
 - c. $\Delta\theta$ = angle θ_f just before the penny slid - angle θ_i when turntable started to rotate
 - d. Δt = time t_f just before the penny slid - time t_i when turntable started to rotate
(use the ω vs. t graph for t_f and vs. t graph for t_i)

5. Repeat steps 2 and 3 to record 3 sets of measurements. Then use cell formulas to determine the mean and standard deviation for each quantity measured.

4. **How many revolutions** Number of revolutions the penny made before sliding off (divide $\Delta\theta$ by 2π)?

5. Use cell formulas to calculate v , a_c , a_t , a_{total} in your data table using the mean measurements.

$$\text{Avg } V = (8.884 \text{ rad/s}^2)(0.05 \text{ m}) = 0.444 \text{ m/s}$$

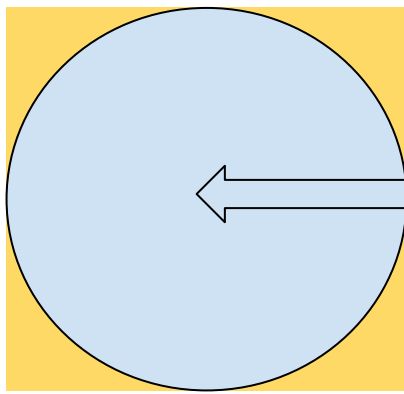
$$\text{Avg } A_c = (8.884 \text{ rad/s}^2)^2(0.05 \text{ m}) = 3.946 \text{ m/s}^2$$

Take aways

1. Uniform vs. Non-uniform circular motion:

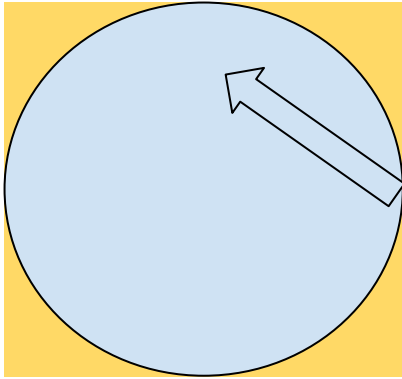
In which direction does friction act on the penny during **uniform circular motion** (constant speed, changing direction)? Draw a vector and describe its direction

The friction acts in the direction pointing towards the center of the system as a centripetal force.



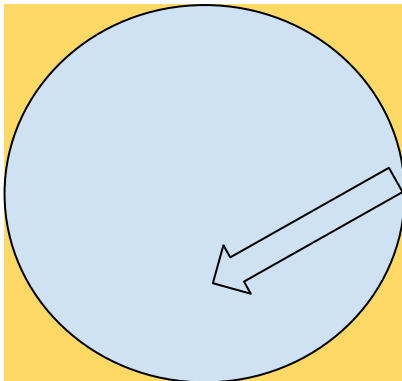
In which direction does friction act on the penny during **non-uniform circular motion speeding up**? Draw a vector and describe its direction (hint: first draw the orthogonal components and add them vectorially).

The friction points forward and inward at an angle as it is the vector sum of a c and a t



In which direction does friction act on the penny during **non-uniform circular motion slowing down**? Draw a vector and describe its direction (hint: first draw the orthogonal components and add them vectorially).

The friction points backwards and inward as it is the vector sum of a_c and a_t



2. Resultant Acceleration:

Write an expression for the resultant (total) acceleration in terms of its radial and tangential components. What has to equal zero for the resultant acceleration to be purely centripetal?

$$a_{total} = \sqrt{a_r^2 + a_t^2}$$

3. Sliding Condition:

Write a statement for when the penny slips in terms of net force, total acceleration and maximum static friction.

The maximum static friction has been reached causing the penny to no longer be supported or be held by the force of friction. Net force was initially zero but once it slips it is no longer zero

causing the penny to move into the non net zero force direction. The acceleration that was once zero is no longer zero when the penny slips, the penny accelerates in the direction away from the spinning table.

Activity 1 determine the coefficient of static friction between the penny and the turntable

ACTIVITY 2 DATA

run #	ω_1 (rad/s)	α (rad/s ²)	$\Delta\theta$ (radians)	Δt (s)	# of revolutions	v(m/s)	a _c (m/s ²)	a _t (m/s ²)	a _{total} (m/s ²)	μ_s
1	8.639	1.257				0.43195	3.7316	0.06285	3.73214529	0.380442945
2	8.786	1.396				0.43930	3.8597	0.0698	3.860320892	0.393508756
3	9.226	1.278				0.46130	4.2560	0.0639	4.256433479	0.433887205
mean	8.884	1.327				0.44400	3.9460	0.066	3.946	0.4026
stdev	0.306	0.098				0.01530	0.2736	0.0037	0.2736	0.0278

Given	SI Units
M = 20 g	0.02kg
r1 = 5 cm	0.05m
r2 = 10 cm	0.10m